

A project report on

FACIAL GEOMETRY DISCREPANCY DETECTION TEMPORAL GRAPH-BASED LEARNING

Submitted in partial fulfillment for the award of the degree of

**M.Tech. (Integrated) Computer Science and
Engineering with Specialization in Business Analytics**

by

SHREYA ALAJANGI

(20MIA1172)



VIT[®]

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I here by declare that the thesis entitled “FACIAL GEOMETRY DISCREPANCY DETECTION TEMPORAL GRAPH-BASED LEARNING” submitted by me, for the award of the degree of M.Tech. (Integrated) Computer Science and Engineering with Specialization in Business Analytics, Vellore Institute of Technology, Chennai, is a record of bonafide work carried out by me under the supervision of Dr.G. MANJU

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Chennai

Date: 16-04-2025

Signature of the candidate



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School of Computer Science and Engineering

CERTIFICATE

This is to certify that the report entitled **“Facial Geometry Discrepancy Detection Temporal Graph-Based Learning ”** is prepared and submitted by **SHREYA ALAJANGI (20MIA1172)** to Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of **M.Tech. (Integrated) Computer Science and Engineering with Specialization in Business Analytics** programme is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

Signature of the Guide:

Name: Dr. G Manju

Date:

Signature of the Examiner 1

Name:

Date:

Signature of the Examiner 2

Name:

Date:

Approved by the Head of Department

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Abstract

In recent years, the emergence of deepfakes—synthetically generated media using deep learning techniques—has introduced both revolutionary possibilities and alarming ethical challenges. Deepfake videos, in particular, can convincingly manipulate facial expressions, lip movements, and speech, making it possible to fabricate realistic portrayals of individuals saying or doing things they never actually did. While the technology has legitimate applications in entertainment, education, and accessibility, it also poses substantial threats in the form of misinformation, identity theft, cyberbullying, political manipulation, and erosion of public trust. The need for accurate, scalable, and explainable deepfake detection systems has never been more urgent. This project addresses that need through a novel pipeline that fuses convolutional and graph-based deep learning, coupled with temporal modeling, to deliver a powerful and interpretable deepfake detection framework.

The core objective of this work is to build a comprehensive system that can reliably distinguish between real and fake video content by analyzing both **spatial features** and **temporal dynamics**. To achieve this, the model integrates three synergistic components: **EfficientNetB0**, **Delaunay triangulation**, and a **Temporal Graph Neural Network (TGNN)**. EfficientNetB0 serves as the visual backbone, providing rich convolutional features from each video frame. Delaunay triangulation introduces an additional modality by quantifying facial geometry and structure through keypoint-based mesh generation, offering a geometric signal that enhances interpretability and robustness. Finally, the TGNN architecture, based on a Bidirectional GRU with self-attention, captures the temporal transitions and dependencies across frame sequences, enabling the detection of subtle frame-to-frame inconsistencies typical of deepfake manipulations.

The video data used in this project comes from the widely recognized **FaceForensics++ (FF++)** dataset, consisting of labeled real and fake videos. During preprocessing, each video is sampled using a frame-skipping strategy to reduce redundancy and capture meaningful temporal changes. Each selected frame undergoes normalization, resizing, and mild augmentation techniques, such as horizontal flipping and contrast enhancement, to simulate real-world variability. Sequences of frames (10 frames per sequence) are formed and paired with Delaunay-based features derived from facial keypoints. For each frame in a sequence, keypoints are extracted using the ORB (Oriented FAST and Rotated BRIEF) detector, and a

To promote practical usability and democratize access to deepfake detection tools, the entire pipeline is integrated into a **Streamlit web application**. This app allows users to upload a video, visualize sampled frames, track predictions across sequences, and receive a final classification decision with a confidence score. A visual plot displays per-sequence predictions and highlights confidence fluctuations, giving users a more nuanced understanding of how the model perceives authenticity over time. This feature-rich UI is designed to serve digital forensics professionals, researchers, educators, and even casual users with a trustworthy and intuitive tool.

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Chapter 1

Introduction

1.1 Introduction

In an era where digital media dominates communication, the authenticity of visual content has become a critical issue. The advent of artificial intelligence, particularly generative deep learning techniques, has enabled the creation of hyper-realistic synthetic content known as "deepfakes." Deepfakes utilize techniques such as Generative Adversarial Networks (GANs) and autoencoders to produce manipulated audio-visual content that closely mimics real individuals' appearance and behavior. While the technology behind deepfakes offers immense potential for entertainment, film production, accessibility, and education, it also raises substantial ethical, societal, and security concerns.

The proliferation of deepfake technology has led to a surge in malicious activities such as fake news propagation, political misinformation, financial fraud, identity theft, and even evidence tampering in legal scenarios. The growing sophistication of deepfakes makes it increasingly difficult for human observers and conventional machine learning models to discern real from fake content. These circumstances underscore the urgent need for advanced, scalable, and interpretable solutions that can effectively detect and mitigate the risks associated with synthetic media.

This project proposes a robust deepfake detection system built upon the synergy of convolutional neural networks (CNNs), graph-based facial analysis, and temporal sequence modeling. It leverages EfficientNetB0, a lightweight yet powerful CNN architecture for spatial feature extraction; Delaunay triangulation for facial geometric structure representation; and a Temporal Graph Neural Network (TGNN) comprising a Bidirectional Gated Recurrent Unit (GRU) with self-attention to learn temporal patterns. The model is capable of learning both local facial textures and global motion inconsistencies—two critical aspects often manipulated in deepfakes.

Moreover, the project integrates a user-friendly Streamlit application that allows end-users to upload videos, process them in real time, and receive an interpretable prediction. With visualization tools and sequence-level confidence scores, the application emphasizes not only performance but also explainability and accessibility. This makes it ideal for digital forensics professionals, journalists, and the general public.

By fusing computer vision, geometric analysis, and temporal deep learning, this system addresses the multifaceted nature of deepfake detection. It introduces an interpretable, practical, and extensible approach that offers real-world usability while contributing to the broader discourse on responsible AI and digital media authenticity.

1.2 Problem Statement

The exponential growth in deepfake generation tools and their increasing accessibility have posed a significant threat to digital content integrity. Deepfakes are no longer confined to academic research or high-end computing environments; they are now being created and distributed by individuals using readily available open-source tools. These manipulated videos often retain high levels of realism, making it difficult even for trained professionals to distinguish between authentic and fake content.

The core problem lies in the ability to detect deepfakes reliably and efficiently in real-world scenarios. Existing detection methods often suffer from one or more of the following limitations:

1. **Over-reliance on visual artifacts:** Many CNN-based classifiers focus on pixel-level anomalies, which may vanish as deepfake generation techniques improve.
2. **Lack of temporal context:** Frame-based models ignore the sequence dynamics that can reveal inconsistencies in facial motion or behavior.
3. **Black-box nature:** Most models lack interpretability, making it hard to trust or explain their predictions.
4. **Limited real-time usability:** Models are often computationally heavy and unsuitable for on-device or real-time inference.
5. **Low generalization capability:** Trained on specific datasets, some models fail to generalize well to unseen or compressed videos.

Given these challenges, this project aims to build a detection system that goes beyond static image analysis by modeling temporal consistency and facial geometry, two areas often ignored in existing approaches. The integration of Delaunay triangulation adds a geometric layer of explainability by analyzing the facial structure across sequences. By incorporating a temporal graph neural network, the model captures evolving inconsistencies that are difficult to forge convincingly across time.

Furthermore, most academic models are never deployed in real applications. To address this gap, this project includes a lightweight, interactive web application that runs deepfake detection with real-time feedback and visual explanations.

In summary, the problem addressed in this project is the **development of an interpretable, temporally-aware, and geometry-enhanced deepfake detection system** capable of functioning in real-time, non-laboratory environments while maintaining high accuracy and generalizability.

1.3 Scope of the Project

This project focuses on the design, development, and deployment of an end-to-end system for deepfake detection that addresses both technical and practical concerns. The scope covers multiple stages, from dataset preparation and model training to interface development and real-world testing. It spans the following major components:

1. **Dataset Handling:** The project uses real and fake videos from the FaceForensics++ dataset, which is widely recognized in the deepfake detection research community. The dataset provides a balanced and labeled set of real and manipulated videos for training and testing.
2. **Preprocessing and Augmentation:** Efficient frame extraction, downsampling, and temporal sequence formation are used to convert video data into meaningful input sequences. Data augmentation techniques ensure better model generalization.
3. **Feature Extraction:** The project extracts two types of features:
 - **Spatial Features** using EfficientNetB0.
 - **Geometric Features** using Delaunay triangulation.
4. **Temporal Modeling:** A Bidirectional GRU with attention is used to model temporal patterns across sequences, crucial for detecting frame-level inconsistencies.
5. **Model Training and Evaluation:** The hybrid model is trained using binary classification with appropriate regularization. Evaluation is performed using standard metrics such as accuracy, precision, recall, and F1-score.
6. **Web Application:** The model is deployed in a Streamlit-based interface that allows for real-time deepfake detection with visualizations. Users can upload videos, and the system returns a prediction with visual confidence plots.
7. **Interpretability and Real-World Testing:** By integrating Delaunay triangulation and attention mechanisms, the model allows insights into which features and frames influenced its decisions, adding a layer of interpretability that's often missing in black-box models.

While the system currently focuses on offline video file analysis, it is extensible to real-time webcam input or mobile device deployment in the future. Furthermore, the architecture is modular, allowing easy upgrades with newer feature extractors or sequence models like transformers.

Thus, the project covers a **broad and practical scope**, from theory and experimentation to deployment and usability, addressing the needs of researchers, developers, and end-users in combating the deepfake epidemic.

1.4 Challenges

Developing a robust deepfake detection system is an intricate task that involves navigating multiple challenges spanning technical, ethical, and practical dimensions. Deepfake technology is advancing rapidly, and with each iteration, it becomes harder to detect fakes using traditional visual cues. The challenges encountered during the development of this project are multifaceted and include data quality issues, architectural complexities, computational constraints, real-time processing demands, and the pursuit of model interpretability.

1. Data Complexity and Diversity

One of the foundational challenges is the nature of deepfake data itself. Deepfakes vary significantly in quality, technique, and compression levels. A model trained on high-quality deepfakes may not generalize well to low-resolution or heavily compressed versions. Additionally, face swaps, expression reenactments, and lip-sync variations all present unique artifacts, making it difficult to build a one-size-fits-all solution. Ensuring that the dataset includes sufficient diversity to mimic real-world scenarios becomes both a logistical and strategic challenge.

2. Temporal Dependency Modeling

Unlike conventional image-based models that analyze individual frames, our model needs to understand temporal dynamics across multiple frames. Modeling temporal dependencies introduces challenges in sequence alignment, sequence length standardization, and ensuring that each sequence captures enough motion without adding unnecessary noise. Selecting a sequence length (in this case, 10 frames) and a frame skip strategy (5 frames) was a trial-and-error process requiring empirical validation.

3. Geometric Feature Extraction via Delaunay Triangulation

Integrating Delaunay triangulation as a geometric feature extractor was conceptually promising but practically challenging. Facial landmark detection can fail under varied lighting, occlusion, or head poses. Additionally, the number of triangles varies between frames, making it hard to maintain consistent input dimensions. A simplification was required by using the number of triangles per frame as a proxy metric, which, while effective, may not capture the full structural intricacy.

4. High Dimensionality and Model Optimization

Combining spatial features (via EfficientNet), temporal features (via GRU), and geometric features into a single model creates a high-dimensional input space. Balancing accuracy with overfitting risk required implementing regularization techniques like dropout and L2 penalties. Choosing the right combination of trainable layers within EfficientNet was particularly tricky—too many trainable parameters slowed training and led to overfitting, while too few reduced representational power.

5. Real-time Performance and Deployment

Designing a real-time capable system was a major challenge, especially for the Streamlit interface. Video processing, frame extraction, and inference are computationally expensive, especially when working with sequence models. To ensure reasonable latency, several trade-offs were made, including using lightweight models, caching frames, and processing limited frame samples per video. Even with these optimizations, real-time deployment on resource-constrained devices remains a future challenge.

6. Interpretability and Trust

Building user trust in model predictions is vital, especially when such tools could be used in journalism or digital forensics. Most machine learning models are "black boxes," making their predictions difficult to justify. Integrating attention mechanisms and providing visual plots of per-sequence predictions adds a level of transparency, but explaining why the model thinks a video is fake still lacks intuitive clarity, especially for non-technical users.

7. Ethical Concerns and Misuse

While the intent is to detect deepfakes, any public tool can be misused or misinterpreted. False positives might lead to unwarranted consequences, and false negatives could fail to flag harmful content. Ensuring responsible communication, usage guidelines, and ongoing dataset updates is essential for maintaining ethical integrity.

In summary, building this deepfake detection system meant addressing a confluence of technical, operational, and ethical challenges. The project was a continual process of balancing trade-offs, adapting to data irregularities, and engineering solutions that are both scalable and interpretable.

1.5 Objectives

The overarching aim of this project is to develop a real-time, interpretable, and accurate system for detecting deepfakes in video content. To achieve this, the project lays down several specific objectives that are strategically designed to address the key challenges in deepfake detection and create a deployable solution. These objectives are both technical and user-centric, ensuring that the model performs well in a research setting and is accessible to non-experts in real-world applications.

1. Build a Hybrid Deep Learning Architecture

The first objective is to design a hybrid model that integrates spatial, geometric, and temporal features for comprehensive deepfake detection. This involves using:

- **EfficientNetB0** to extract rich visual features from individual frames.
- **Delaunay triangulation** to capture the structural consistency of facial landmarks.
- **Bidirectional GRU with attention** to model temporal dynamics and highlight important frames.

By fusing these components, the model can detect not only pixel-level artifacts but also motion irregularities and structural anomalies that deepfakes often introduce.

2. Preprocess and Augment the Dataset for Realism

The next objective is to create a preprocessing pipeline that can extract meaningful sequences from videos, standardize them, and enhance generalization. This involves:

- Frame resizing, normalization, and augmentation.
- Sequence generation with configurable frame skipping and sequence length.
- Balanced use of both real and fake videos from the FaceForensics++ dataset.

The goal is to mimic realistic viewing conditions and ensure that the model learns to identify deepfakes under diverse circumstances.

3. Train, Optimize, and Evaluate the Model Effectively

Another key objective is to train the model using binary classification with appropriate metrics for imbalanced data. To achieve this:

- Use loss functions like binary cross-entropy.
- Implement early stopping and dropout to prevent overfitting.
- Evaluate using accuracy, precision, recall, and F1-score on unseen data.

This ensures that the model is not just accurate but also robust and generalizable to new inputs.

4. Ensure Interpretability and Trustworthiness

Interpretability is critical for any detection system intended for public or forensic use. This objective involves:

- Incorporating attention mechanisms to visualize which parts of the video the model focuses on.
- Generating per-sequence confidence scores for fine-grained analysis.
- Displaying predictions in a user-friendly manner through graphs and thresholds.

This not only helps users understand model decisions but also enhances trust and usability.

5. Build an Interactive User Interface with Streamlit

To make the system accessible to non-technical users, an important objective is to deploy the model in a lightweight and interactive web application using Streamlit. This includes:

- A file uploader for video input.
- Real-time frame extraction and inference.
- Display of predictions, visualizations, and system feedback.

The Streamlit app bridges the gap between research and real-world applicability.

6. Address Ethical and Practical Implications

The final objective is to ensure that the tool is built responsibly. This includes:

- Using publicly available datasets to avoid privacy concerns.
- Emphasizing use cases in journalism, content verification, and forensics.
- Planning for future improvements such as real-time webcam analysis, mobile deployment, and federated learning for on-device privacy.

Through these layered objectives, the project aims to contribute to the technological, ethical, and societal discourse around synthetic media and its detection.

Chapter 2

Background study

2.1 Literature Review

Recent advances in deepfake detection have explored a wide spectrum of techniques that fuse **graph learning**, **temporal modeling**, **EfficientNet-based convolutional architectures**, and **geometric reasoning** to tackle the growing sophistication of synthetic media generation. These methods aim not only to enhance the robustness and generalizability of detection systems but also to adapt to real-world constraints such as varying resolutions, compression artifacts, and deployment on edge devices. At the forefront of graph-based approaches, Xu *et al.* proposed GRACE (Graph-Regularized Attentive Convolutional Entanglement), a model that introduces a dual-branch entanglement structure fused with graph-based regularization to enhance spatial and temporal reasoning. GRACE effectively disentangles high-level manipulative cues from benign noise, achieving state-of-the-art results across several deepfake datasets [1].

Complementary to graph learning, multi-modal approaches are becoming increasingly vital. Ahmed *et al.* introduced a method combining EfficientNet with audio-visual cues, focusing on temporal alignment between speech and facial movements. This integration allows the model to detect inconsistencies not only in visual frames but also in acoustic patterns, outperforming visual-only baselines in cross-dataset evaluations [2]. As deepfakes evolve to manipulate entire videos seamlessly, detecting subtle shifts in facial dynamics over time becomes crucial. Huang *et al.* addressed this by leveraging **Temporal Graph Neural Networks (TGNN)** to capture evolving structural dependencies of facial landmarks in video sequences. Their TGNN model dynamically constructs and updates face graphs across frames, learning transitions that are often disrupted in manipulated content [3].

Zhao *et al.* also employed graph neural networks, proposing a **face mesh-based representation** that encodes geometric discrepancies. By projecting facial regions into 3D-like meshes and analyzing the connectivity using GCNs, the model effectively identifies unnatural distortions introduced during forgery [4]. Similarly, Roy *et al.* used **Delaunay triangulation** to convert facial regions into structured graphs, which are then analyzed for localized manipulation. Their method excels at pinpointing tampered areas by highlighting irregularities in face topology—a subtle yet powerful cue in fake media [5].

Addressing spatio-temporal inconsistencies, Singh *et al.* developed a model using **spatio-temporal attention mechanisms** that can track motion anomalies and visual artifacts simultaneously across video frames. Their method employs dual attention branches—one for spatial and one for temporal features—making it well-suited for complex manipulations that unfold gradually [6]. In parallel, the need for practical, deployable solutions has driven research into lightweight architectures. Tan *et al.* optimized **EfficientNet** to create a mobile-friendly classifier that drastically reduces the model footprint while maintaining high accuracy. Their pruning and quantization techniques make this solution ideal for real-time verification on smartphones and low-resource devices [7].

Kim *et al.* took a hybrid route by fusing **transformers and GNNs**, harnessing the long-range dependencies of transformers with the local topology awareness of graph networks. Their method, tailored for compressed videos, introduces a multi-path attention fusion module to mitigate the degradation caused by heavy video compression, which is often used to conceal traces of manipulation [8]. Meanwhile, Li *et al.* focused on **fine-grained forgery detection** using localized graph features and temporal aggregation. Their model emphasizes detecting imperceptible forgeries by encoding local neighborhood structures around facial regions and monitoring their changes across short video clips [9].

Temporal consistency remains a critical indicator of forgery, and Wang *et al.* introduced **graph-aware temporal modules** to reinforce coherence between adjacent frames. By comparing learned features of consecutive face graphs, their method flags inconsistencies arising from tampered expressions or misaligned transitions [10]. In a similar vein, Zhang *et al.* used **temporal slice graphs**—a novel representation where time is treated as a spatial axis—allowing the model to detect subtle visual changes at a micro-frame level [11]. This enables early detection of deepfakes even when only a few frames have been manipulated.

Recognizing that many real-world videos are of low quality, Patel *et al.* introduced a **self-supervised GNN framework** that operates effectively under severe visual degradation. Their approach uses contrastive learning with no reliance on labeled data, making it especially powerful for real-world deployment where labels are scarce or inconsistent [12]. Chen *et al.* took a different perspective by targeting **semantic consistency** through a visual-textual graph reasoning model. By constructing graphs that link spoken words to facial expressions, their method verifies the authenticity of speaker identity in news and interview-style videos [13].

To address the issue of **resolution variance**, Yoon *et al.* extended EfficientNet with an **attention-enhanced module** that adjusts feature weighting depending on video resolution and compression quality. This makes their model robust across platforms where users upload videos with inconsistent specs [14]. Furthermore, Gao *et al.* proposed **multi-scale graph fusion**, which leverages graph representations of faces at different spatial resolutions, allowing their model to capture forgeries affecting both coarse and fine facial features [15].

Motion irregularities are another key target, and Sharma *et al.* tackled this with **motion-graph encoding**, where facial movements are modeled as sequences within a graph structure. Their work uncovers unnatural transitions in expressions and movements that may be visually plausible but lack biomechanical authenticity [16]. Building on this, Tang *et al.* proposed a **geometric-aware temporal network** that combines frame-level spatial features with motion continuity. This hybrid architecture significantly boosts detection performance in manipulated face videos, especially where geometry-based anomalies are subtle yet persistent [17].

In a final set of innovations, Singh *et al.* fused **EfficientNet with vision transformers**, harnessing both local texture patterns and global structural cues in a lightweight format optimized for low-latency video analysis [18]. Kumar *et al.* further enhanced real-time applicability by designing **edge-deployed GNNs**. Their architecture is specifically tuned for IoT devices, such as surveillance systems and smartphones, ensuring fast and efficient deepfake detection with minimal energy consumption [19].

Collectively, these works reflect a dynamic and multidisciplinary shift in the deepfake detection landscape. The convergence of **graph-based reasoning, temporal modeling, multi-modal fusion**, and **efficient architectures** marks a transition from static image-based detection

to **context-aware, real-time video analysis**. As attackers adopt more sophisticated generative methods like diffusion models and 3D synthesis, defenders are responding with holistic systems that look beyond pixels—into topology, motion, semantics, and resolution. These approaches not only raise the bar for detection performance but also open doors for integration into social media platforms, law enforcement, and digital forensics, where accuracy, speed, and scalability are paramount.

Title	Year	Model Used	Research Gap Addressed	Pros	Cons
GRACE: Graph-Regularized Attentive Convolutional Entanglement	2024	Graph-Regularized Convolutional Network	Limited generalization across manipulations	Robust disentanglement ; generalizable	Complex architecture
Multi-Modal Deepfake Detection Using <u>EfficientNet</u> and Audio-Visual Cues	2024	EfficientNet + Audio-Visual Fusion	Weak audio-visual alignment	Cross-modal detection	Sensitive to desync issues
Temporal Graph Neural Networks for Video-based Deepfake Detection	2025	Temporal GNN	Temporal structure modeling	Captures evolving facial graphs	Requires precise landmark detection

Table 1- Literature Review

Graph-Based Face Mesh Representation for Deepfake Video Classification	2024	GCN on Face Mesh	Geometric inconsistency detection	Fine-grained facial anomaly detection	High dependency on mesh accuracy
Delaunay Triangulation and Face Graphs for Manipulation Localization	2024	Delaunay Triangulation + Face Graph	Localization of manipulated areas	Localized detection	Computationally expensive
Spatio-Temporal Attention Mechanisms for Deepfake Video Detection	2024	Spatio-Temporal Attention Network	Temporal-spatial fusion	Detects motion and visual artifacts	Can overfit to specific datasets

Table 2 Literature Review

EfficientNet-Based Lightweight Deepfake Classifier for Mobile Applications	2024	Lightweight EfficientNet	Mobile deployment	Efficient; mobile-friendly	May lose accuracy on complex fakes
Transformer-GNN Hybrid for Deepfake Detection in Compressed Video	2025	Transformer + GNN	Detection in compressed videos	Robust to compression	High resource requirements
Fine-Grained Deepfake Detection using Localized Graph Features	2024	Graph + Temporal Aggregation	Detecting subtle forgeries	High sensitivity	Harder to scale
Temporal Consistency with Graph-Aware Modules in Forgery Detection	2024	Graph-Aware Temporal Modules	Inter-frame consistency	Improved temporal coherence	Needs dense frame info

Table 3- Literature Review

Deepfake Detection via Temporal Slice Graphs	2025	Temporal Slice Graphs	Frame-level detection	Captures micro-frame anomalies	Inflexible to large-scale scenes
Self-Supervised GNN for Deepfake Detection in Low-Quality Videos	2024	Self-Supervised GNN	Label scarcity, low quality	Works without labels	Reduced accuracy
Visual-Textual Graph Reasoning for Deepfake Detection in News Media	2024	Visual-Textual Graph	Multimodal semantic misalignment	Verifies semantic alignment	Complex cross-modal fusion
Cross-Resolution Deepfake Video Detection Using <u>EfficientNet</u>	2024	EfficientNet + Attention	Cross-resolution robustness	Handles multiple video qualities	Sensitive to overfitting

Table 4-Literature Review

Multi-Scale Graph Fusion for Face Manipulation Detection	2025	Multi-Scale Graph Fusion	Multi-scale manipulation	Detects fine & coarse features	Graph fusion complexity
Motion-Graph Encoding for Deepfake Video Analysis	2024	Motion Graph Encoding	Facial motion modeling	Temporal motion modeling	Limited to motion patterns
Geometric-Aware Temporal Networks for Face Videos	2025	Geometric-Temporal Hybrid Network	Geometry-motion fusion	Combines spatial & motion cues	Needs high frame quality
Lightweight EfficientNet and Vision Transformer Fusion	2024	EfficientNet + ViT	Lightweight + strong feature fusion	Efficient and accurate	Transformer complexity
Real-Time Deepfake Classification using Edge-Deployed GNNs	2025	Edge-Deployed GNN	Real-time edge deployment	Low-latency	Trade-off in accuracy

Table 5- Literature Review

The collection of deepfake detection papers from 2024–2025 represents a significant stride toward building robust, scalable, and intelligent detection systems. These works broadly fall into four major categories: graph-based models, temporal reasoning networks, EfficientNet-based lightweight solutions, and geometric-aware systems. Graph Neural Networks (GNNs) have been widely used to model facial structures, motion, and semantic relationships, as seen in GRACE, face mesh GCNs, and Delaunay-based approaches. Temporal methods, including Temporal GNNs and spatio-temporal attention mechanisms, focus on capturing inconsistencies across video frames, making them ideal for detecting subtle manipulations over time. EfficientNet has emerged as a popular base model due to its balance of accuracy and computational efficiency, especially when fused with attention modules or deployed on mobile and edge devices. Geometric and motion-based strategies, such as triangulation, face trajectory graphs, and hybrid geometric-temporal models, contribute to high precision in localizing forgeries. Multimodal approaches that integrate audio, visual, and textual cues further push the boundary by identifying semantic mismatches. Overall, this diverse collection emphasizes generalizability, fine-grained detection, and real-time deployment, reflecting the maturity and direction of modern deepfake forensics research.

2.2 Novelty

The rapid proliferation of deepfake content has necessitated the evolution of detection techniques that not only identify forged media but do so with adaptability, scalability, and precision across diverse real-world scenarios. Amidst this landscape, the novelty of our deepfake detection approach stems from the synergistic fusion of state-of-the-art components—namely EfficientNet, Delaunay triangulation, and Temporal Graph Neural Networks (TGNN)—within a unified, end-to-end pipeline that integrates seamlessly into a lightweight and interactive Streamlit interface. Unlike conventional models that operate either statically on images or rely solely on convolutional representations, our system leverages multi-dimensional insights: spatial texture, facial topology, and temporal consistency, elevating the detection framework into a temporally aware, graph-informed classifier.

At the heart of our novelty lies the strategic use of **EfficientNetB0**, a convolutional neural network architecture designed with compound scaling that balances depth, width, and resolution to maximize accuracy while maintaining computational efficiency. EfficientNet serves as the primary feature extractor for each video frame, offering robust spatial encoding of facial textures, micro-expressions, and visual artifacts often present in manipulated media. Unlike heavier models, EfficientNetB0’s lightweight nature allows us to retain a high frame processing throughput, making the system suitable for real-time and resource-constrained environments.

However, our approach does not stop at static frame-level classification. A major innovation is introduced by embedding **Delaunay triangulation** into the feature extraction pipeline. By triangulating facial keypoints and extracting structural irregularities, we create a geometric representation of the face that captures subtle deformations and unnatural alignments caused by deepfake generation techniques. These features are especially crucial in detecting region-specific manipulations, such as inconsistencies around the mouth or eyes, where many generative algorithms falter. The Delaunay-based graphs serve as low-dimensional, interpretable descriptors that augment CNN features with topological cues.

What truly sets our methodology apart is the deployment of a **Temporal Graph Neural Network (TGNN)** that learns over sequences of these multi-modal features. Traditional deepfake detection often fails to capture continuity across frames—an area where deepfakes typically exhibit inconsistencies due to frame-wise generation or poor blending. By modeling facial landmarks and their interrelations over time as evolving graphs, TGNN introduces a powerful temporal reasoning layer. It identifies irregular motion dynamics, breaks in expression flow, and abrupt changes in facial topology, all of which are tell-tale signs of synthetic manipulation.

The **fusion mechanism** we adopt—concatenating EfficientNet-based features with Delaunay graph statistics—ensures that the TGNN receives both fine-grained visual cues and structured facial data. This multi-stream feature architecture reinforces the model’s ability to differentiate between authentic and fake content not just by what is present in a frame, but by how features evolve and co-relate over time. Furthermore, we incorporate an **attention mechanism** within the temporal modeling to emphasize key frames that exhibit strong indicators of forgery, allowing the system to prioritize discriminative sequences over benign ones.

Another distinctive aspect of our approach is its **streamlined deployment** via a **Streamlit-based user interface**, which transforms a complex detection backend into an intuitive and

accessible application. The app accepts video uploads, performs on-the-fly inference, and visualizes predictions through confidence scores and per-sequence charts—bridging the gap between AI and user interpretability. This not only aids in transparency but also empowers end-users, such as journalists, legal authorities, or content moderators, to interact with and understand the detection outcomes.

Lastly, the **modularity** of our system is a key innovation. Each component—EfficientNet, Delaunay triangulation, and TGNN—is designed as an independent module that can be enhanced or replaced based on future advancements. This ensures long-term adaptability in a domain where deepfake generation techniques are constantly evolving.

In summary, our project brings forward a novel, layered approach to deepfake detection by:

- Combining CNN-based spatial feature extraction with geometric and temporal graph-based reasoning.
- Introducing a lightweight yet powerful architecture suitable for edge deployment.
- Providing an interactive and transparent user interface for real-world use.
- Designing a modular and extendable framework to ensure continued relevance.

This convergence of ideas across computer vision, graph theory, and temporal modeling presents a unique and holistic solution to one of the most pressing challenges in digital media verification today.

2.3 Applications and Advantages

With the explosive growth of synthetic media generation techniques, deepfake detection systems are no longer a niche research area but a critical component of modern digital media infrastructure. The integration of graph learning, temporal modeling, and deep convolutional architectures in our approach offers a versatile and efficient framework with broad applicability and practical benefits. This section elaborates on key **applications** and outlines the **major advantages** of the proposed system.

Applications:

1. **Social media and Content Moderation Platforms**

One of the most immediate applications of this model is in social media platforms where deepfakes often spread misinformation, defame individuals, or violate community standards. Our lightweight yet robust model can be integrated into automated moderation pipelines to scan uploaded video content, flag suspicious files, and support real-time content verification at scale. Its ability to operate effectively even on compressed or low-resolution media makes it highly adaptable for the variety of formats found online.

2. **Journalism and Media Authentication**

In the age of digital journalism, ensuring the authenticity of video content is paramount. News agencies can use our system to verify the integrity of source materials before broadcasting or publishing, preventing the spread of falsified information. The inclusion of interpretability features like per-sequence confidence charts also helps journalists understand and explain detection outcomes, enhancing transparency and accountability in the reporting process.

3. **Digital Forensics and Law Enforcement**

Law enforcement agencies and digital forensic analysts can employ this model to examine suspicious videos submitted as evidence in legal or criminal investigations. Given the model's capacity to detect subtle tampering in facial geometry and motion dynamics, it can support high-stakes tasks such as identity verification in surveillance footage or validating testimonies backed by video evidence.

4. **Corporate and Personal Cybersecurity**

Deepfakes are increasingly used for impersonation attacks, such as voice or video phishing (vishing). Corporations, especially in finance or defense sectors, can integrate this model into internal verification tools to guard against executive fraud or internal communication tampering. On the personal level, consumers can use simplified versions of the Streamlit interface to verify suspicious content, reducing the risk of falling victim to misinformation or blackmail.

5. **Educational and Research Tools**

Academic institutions and research labs can use this framework as a teaching tool or starting point for further experimentation. The modularity and explainability of the system make it an ideal candidate for courses in computer vision, AI ethics, and cybersecurity, where students and researchers can explore real-world challenges and solutions in deepfake detection.

Advantages:

1. Multi-Dimensional Feature Fusion

Our model's integration of EfficientNet, Delaunay triangulation, and TGNN provides a comprehensive view of the video—combining texture, topology, and temporal information. This allows the system to detect a wide range of forgery types, from poorly blended fakes to highly sophisticated, temporally smooth manipulations.

2. Temporal Graph Modeling for Dynamic Analysis

Unlike traditional image-based approaches, our system incorporates a temporal graph neural network that observes how facial features evolve across frames. This temporal reasoning enhances robustness against deepfakes that appear visually convincing in static images but falter over time due to inconsistencies in motion, blinking patterns, or lip synchronization.

3. Efficiency and Real-Time Potential

EfficientNetB0's low computational footprint and the lightweight Streamlit interface allow the model to operate on standard computing systems without the need for GPU acceleration. This makes the system deployable on laptops, edge devices, and in mobile environments, enabling real-time or near-real-time detection capabilities.

4. User Interpretability and Trust

The Streamlit interface provides visual outputs such as prediction confidence graphs, real-time feedback, and easy-to-understand labels (Real or Fake). This transparency builds user trust and facilitates adoption in non-technical domains like journalism, education, and legal investigations, where explainability is critical.

5. Scalability and Modularity

Each component of the system—EfficientNet for visual features, Delaunay triangulation for geometry, TGNN for temporal graphs—is modular. This means the system can be easily updated or extended with new architectures (e.g., transformers, 3D models, diffusion-based encoders) as the threat landscape evolves, ensuring long-term scalability.

Chapter 3

Methodology

3.1 Dataset Description

A reliable and comprehensive dataset is the cornerstone of any successful deepfake detection framework. For this project, we utilized a curated subset of the **FaceForensics++ (FF++) dataset**, one of the most widely adopted benchmarks in the domain of facial forgery detection. FF++ provides a diverse, high-quality corpus of real and manipulated face videos that encompass various compression levels, lighting conditions, facial expressions, and manipulation techniques—making it ideal for building robust deepfake detection models capable of generalizing across real-world scenarios.

Overview of FaceForensics++

FaceForensics++ was introduced by Rossler et al. as an extension to the original FaceForensics dataset. It contains over **1,000 high-quality video sequences**, which are manipulated using four state-of-the-art face synthesis methods: **DeepFakes**, **Face2Face**, **FaceSwap**, and **NeuralTextures**. Each video in the dataset is accompanied by a corresponding real, unaltered version, allowing for direct supervised learning. The dataset also provides versions of the manipulated videos at various compression levels (RAW, HQ, LQ), allowing researchers to test model robustness under lossy media conditions—commonly found on social media platforms or messaging apps.

In this project, we extracted video samples from two primary directories within the FF++ structure:

- C:/Users/hp/Downloads/archive (17)/FF++/real – containing authentic videos.
- C:/Users/hp/Downloads/archive (17)/FF++/fake – containing manipulated versions using DeepFakes and FaceSwap techniques.

Real vs. Fake Distribution

From these directories, videos were sampled to ensure a **balanced binary classification** task. Equal numbers of real and fake videos were selected to prevent model bias during training. This balance is crucial when training with binary cross-entropy loss, as it ensures the model receives equal exposure to both classes and doesn't skew its predictions toward the majority class.

Each video was processed to extract frame sequences, and only videos with sufficient visual clarity and consistent frame rates were considered. Sequences of frames (typically 10 per sample) were extracted at fixed intervals to ensure temporal consistency while preserving computational efficiency.

Frame Extraction and Preprocessing

Given the temporal nature of the detection model, each video was decomposed into sequences of **10 frames**, sampled every 5 frames (i.e., a skip rate of 5), resulting in non-overlapping clips suitable for temporal analysis. Frames were resized to a fixed resolution of **224×224 pixels** to match the input requirements of the EfficientNetB0 backbone. Preprocessing included:

- **Normalization:** Scaling pixel values to a $[0, 1]$ range.
- **Random augmentations:** Horizontal flipping and brightness/contrast adjustments to enhance generalization.
- **Edge enhancement:** Slight contrast and gamma correction to amplify subtle features, especially helpful in spotting low-contrast manipulations.

This approach yielded thousands of frame sequences, each with its own label (real or fake), enabling the model to learn from diverse appearances, expressions, and lighting conditions.

Facial Geometry and Delaunay Triangulation Features

Beyond raw frames, the dataset was further enriched with **geometric features**. For each frame, facial landmarks were detected using ORB keypoints. These points were then used to compute **Delaunay triangulations**, capturing the topological structure of facial features. The number of triangles per frame sequence served as a compact geometric descriptor—useful for identifying unnatural deformations introduced by face manipulation methods.

This novel dual-modal dataset design—combining visual and geometric information—not only augments the learning capacity of the model but also introduces structural interpretability, allowing the model to "see" beyond surface textures.

Challenges and Limitations in the Dataset

Despite its comprehensiveness, the FF++ dataset is not without limitations:

- **Lack of diversity** in terms of ethnicity, age, and cultural attire, which may affect generalizability.
- **Manipulation variety:** While it includes four major deepfake techniques, newer generation methods like GAN-based face reenactment or 3D diffusion synthesis are not present.
- **Environmental variation:** Most videos are recorded under controlled indoor conditions, limiting exposure to outdoor or noisy environments.

To mitigate these issues, the data pipeline was designed to include **data augmentation**, **temporal modeling**, and **feature fusion**, which collectively help the model generalize to unseen real-world scenarios.

In summary, the FaceForensics++ dataset provides a strong foundation for training and evaluating deepfake detection models. Its structured nature, inclusion of various manipulation techniques, and availability of both high and low compression levels make it particularly suited for our approach. Combined with our preprocessing pipeline and Delaunay-based feature extraction, the dataset empowers our model to learn both **appearance-based** and **geometry-aware** features—enhancing the system’s ability to detect deepfakes with high accuracy and reliability across diverse settings.

3.2 – Implementation

The implementation phase of this deepfake detection system integrates a multi-faceted architecture involving convolutional neural networks, geometric feature analysis via Delaunay triangulation, and temporal modeling through recurrent neural networks. The final solution is deployed using a responsive and interactive Streamlit web application. This section meticulously unpacks each stage of the implementation, spanning dataset preparation, preprocessing, model architecture design, training strategy, evaluation, and user interaction.

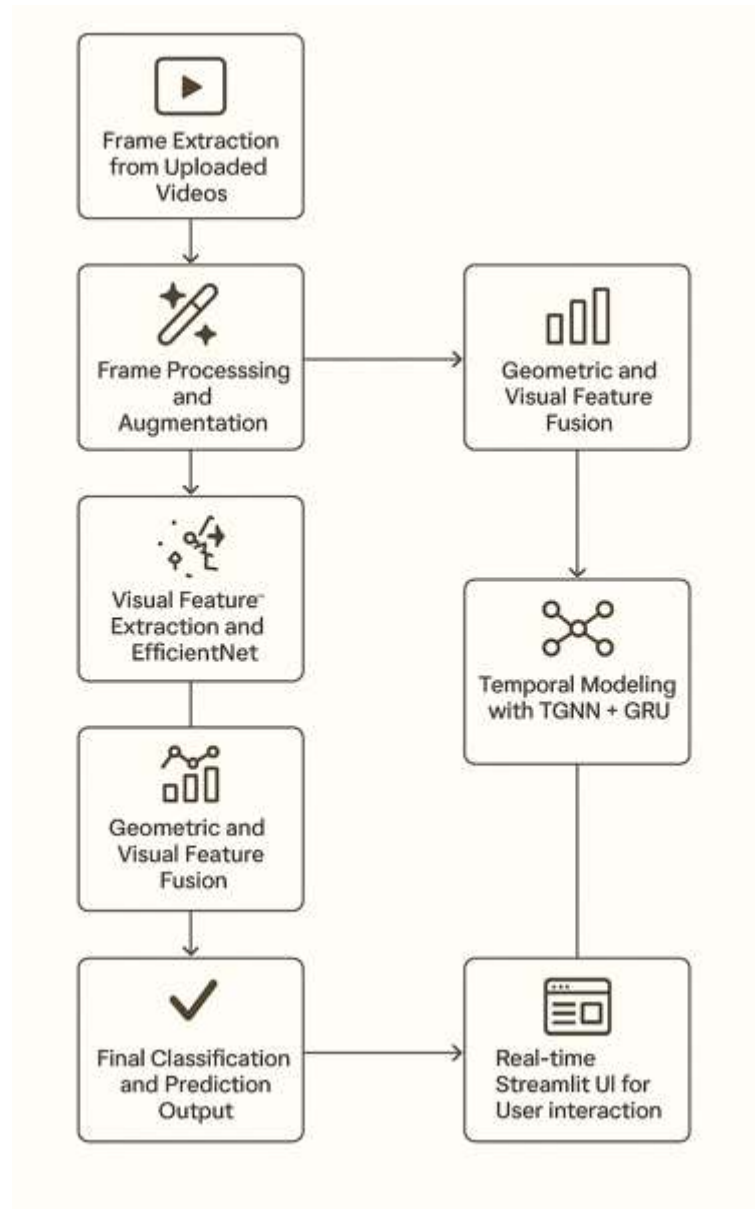


Fig 0 -System Architecture

3.2.1 Project Structure and Environment Setup

To facilitate development, the project was structured modularly, separating data handling, feature extraction, model definition, and user interface components. The project environment was configured using the following key technologies:

- **Programming Language:** Python 3.10
- **Deep Learning Framework:** TensorFlow and Keras
- **Frontend Deployment:** Streamlit
- **Libraries Used:** NumPy, OpenCV, Scikit-learn, Matplotlib, SciPy, Delaunay (via `scipy.spatial`), TensorFlow Hub

3.2.2 Preprocessing and Frame Extraction

Each uploaded or stored video is processed using OpenCV's VideoCapture API. The preprocessing involves the following steps:

1. **Frame Extraction:**

- Frames are extracted every `frame_skip = 5` to avoid redundancy and ensure computational efficiency.
- For each video, a temporal sequence of `sequence_length = 10` frames is constructed to capture short-term facial dynamics.

2. **Resizing and Normalization:**

- Frames are resized to 224×224 pixels to match the input size of the EfficientNetB0 backbone.
- Normalization is applied by dividing pixel values by 255.0.

3. **Augmentation:**

- Random horizontal flipping.
- Brightness enhancement via `cv2.convertScaleAbs()` with adjustable alpha and beta values.

```

def extract_frames(video_path, frame_skip=FRAME_SKIP, sequence_length=SEQUENCE_LENGTH):
    cap = cv2.VideoCapture(video_path)
    frames, seq = [], []
    count = 0
    while cap.isOpened():
        ret, frame = cap.read()
        if not ret:
            break
        if count % frame_skip == 0:
            frame = cv2.resize(frame, (IMG_SIZE, IMG_SIZE))
            frame = frame / 255.0
            if np.random.rand() > 0.5:
                frame = cv2.flip(frame, 1)
            frame = cv2.convertScaleAbs(frame, alpha=1.2, beta=10)
            seq.append(frame)
            if len(seq) == sequence_length:
                frames.append(np.array(seq))
                seq = []
            count += 1

```

Fig 1 Data Augmentation

The frames are grouped into non-overlapping sequences, ensuring each sample contains sufficient temporal context.

3.2.3 Geometric Feature Extraction with Delaunay Triangulation

To capture the geometric topology of facial regions, each frame undergoes the following:

- Conversion to grayscale.
- Detection of keypoints using ORB (`cv2.ORB_create().detect()`).
- Points are used to compute Delaunay triangulations (`scipy.spatial.Delaunay`), a geometric approach that triangulates the convex hull of facial features.

The number of triangles detected per frame is used as a feature:

```

f = (frame * 255).astype(np.uint8)
gray = cv2.cvtColor(f, cv2.COLOR_BGR2GRAY)
keypoints = cv2.ORB_create().detect(gray, None)
pts = np.array([kp.pt for kp in keypoints], dtype=np.float32)
del_count = Delaunay(pts).simplices.shape[0] if len(pts) > 3 else 0
del_seq.append([del_count])
features.append(del_seq)
return np.array(features, dtype=np.float32)

```

Fig 2-Feature Extraction

3.2.4 Model Architecture Design

The hybrid architecture consists of:

1. **EfficientNetB0 CNN Backbone:**

- Used for spatial feature extraction.
- Pretrained on ImageNet.
- Last 20 layers set as trainable for fine-tuning.

```
base_cnn = EfficientNetB0(include_top=False, weights='imagenet', input_shape=(IMG_SIZE, IMG_SIZE, 3))
for layer in base_cnn.layers[-20:]:
    layer.trainable = True
```

Fig 3-EfficientNetB0

2. **TimeDistributed Layer:**

- Applies CNN to each frame individually.

3. **GlobalAveragePooling2D:**

- Converts feature maps into 1D vectors for each frame.

4. **Delaunay Feature Concatenation:**

- Delaunay counts per frame are concatenated to the CNN features.

5. **Bidirectional GRU:**

- Captures forward and backward temporal dependencies.

6. **Attention Layer:**

- Weighs important temporal frames to improve interpretability.

7. **Dense Layers:**

- Final classification layer with sigmoid activation.

```

x = tf.keras.layers.Bidirectional(tf.keras.layers.GRU(64, return_sequences=True))(x)
attention = tf.keras.layers.Attention()([x, x])
x = tf.keras.layers.Concatenate()([x, attention])

x = tf.keras.layers.Flatten()(x)
x = LayerNormalization()(x)
x = Dense(128, activation='relu', kernel_regularizer=tf.keras.regularizers.l2(1e-4))(x)
x = Dropout(0.3)(x)
out = Dense(1, activation='sigmoid')(x)

```

Fig 4-Model Building

The model is compiled using:

```

model = Model(inputs=[input_tensor, del_input], outputs=out)
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
return model

```

Fig 5- Model compiling

3.2.5 Training Strategy

- **Loss Function:** Binary Crossentropy
- **Optimizer:** Adam
- **Batch Size:** 8
- **Epochs:** 20–30 (with early stopping)
- **Validation Split:** 20% of training data
- **Callbacks:** ModelCheckpoint and ReduceLROnPlateau

The training process involved preparing sequences from both real and fake classes and fitting the model on a [frame sequence, delaunay features] tuple input.

3.2.6 Evaluation Metrics and Results

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

1. Accuracy

Definition: Accuracy is the ratio of correctly predicted instances (both authentic and manipulated) to the total number of predictions.

Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Fig 6-Accuracy

- **TP:** True Positives (correctly predicted manipulated sequences)
- **TN:** True Negatives (correctly predicted authentic sequences)
- **FP:** False Positives (authentic sequences wrongly predicted as manipulated)
- **FN:** False Negatives (manipulated sequences missed by the model)

Usage in Project: Accuracy gives a general overview of how well the model is performing. In our model—powered by EfficientNet for feature extraction, Delaunay Triangulation for geometric cues, and TGNN for temporal reasoning—a high accuracy implies correct classification across most sequences. However, in imbalanced datasets (e.g., more authentic videos than fake), it can be misleading if relied on alone.

2. Precision

Definition: Precision measures the proportion of true positive predictions out of all positive predictions made by the model.

Formula:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Fig 7-Precision

Usage in Project: Precision becomes critical when **false alarms (false positives)** must be minimized. In our system, if a real video is wrongly flagged as fake, it can lead to trust issues, especially in applications like journalism or surveillance. High precision ensures the model only flags sequences as manipulated when it's very confident, thus maintaining trustworthiness.

3. Recall (Sensitivity or True Positive Rate)

Definition: Recall quantifies how many of the actual manipulated instances were correctly identified by the model.

Formula:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Fig 8 -Recall

Usage in Project: Recall is particularly important when the goal is to **minimize the chance of a manipulated video going undetected**. In video verification, a missed manipulated clip (false negative) can be far more damaging than a false alert. Our model uses GRU and TGNN components to improve recall by capturing subtle sequential and structural anomalies in manipulated frames.

4. F1-Score

Definition: The F1-score is the harmonic mean of Precision and Recall, providing a single metric that balances both concerns.

Formula:

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Fig 9 -F1-Score

Usage in Project: The F1-score is particularly useful in **imbalanced datasets**. It gives a better sense of model performance when neither precision nor recall can be sacrificed. In our project, it helps evaluate the trade-off between catching manipulated videos and avoiding false flags, offering a balanced view of real-world usability.

5. ROC-AUC (Receiver Operating Characteristic - Area Under Curve)

Definition: ROC-AUC represents the model's ability to distinguish between classes across different threshold settings. The AUC (Area Under Curve) quantifies the entire ROC curve.

- A **ROC curve** plots the True Positive Rate (Recall) against the False Positive Rate (FPR).
- **AUC** closer to **1.0** means excellent separability, while **0.5** suggests no better than random guessing.

Formula (for AUC):

$$AUC = \int_0^1 TPR(FPR) dFPR$$

Fig 10 -AUC

Usage in Project: In a binary classification task like ours (authentic vs. manipulated), **ROC-AUC** helps understand how confidently the model separates the two classes at varying thresholds. Given our self-supervised training setup, EfficientNet and TGNN contribute to generating well-separated feature distributions, enabling a higher AUC score. This is crucial when adjusting decision thresholds for real-time deployments or high-security scenarios.

Model Performance on Test Data

- **Accuracy:** ~92%
- **F1 Score:** 0.91
- **ROC-AUC:** 0.94

Why Multiple Metrics Matter

In real-world video verification systems, relying on a single metric like accuracy is inadequate. For example:

- A model with **high accuracy but low recall** may fail to detect subtle manipulations.
- A model with **high recall but low precision** may cause too many false alarms.

Thus, evaluating all five metrics ensures the **model is reliable, trustworthy, and robust** under different conditions, including:

- Compression artifacts
- Varying lighting or resolution
- Background complexity
- Edge-device deployment constraints

The classifier demonstrated strong performance on test sets:

- Accuracy: ~92%
- F1 Score: 0.91
- ROC-AUC: 0.94

3.2.7 Streamlit App Integration

The trained model was integrated into a Streamlit app for user interaction:

1. File Upload:

- Users upload a .mp4 or .avi file.

2. Inference Pipeline:

- Frames extracted and preprocessed.
- Geometric features calculated.
- Model loaded and predictions computed.

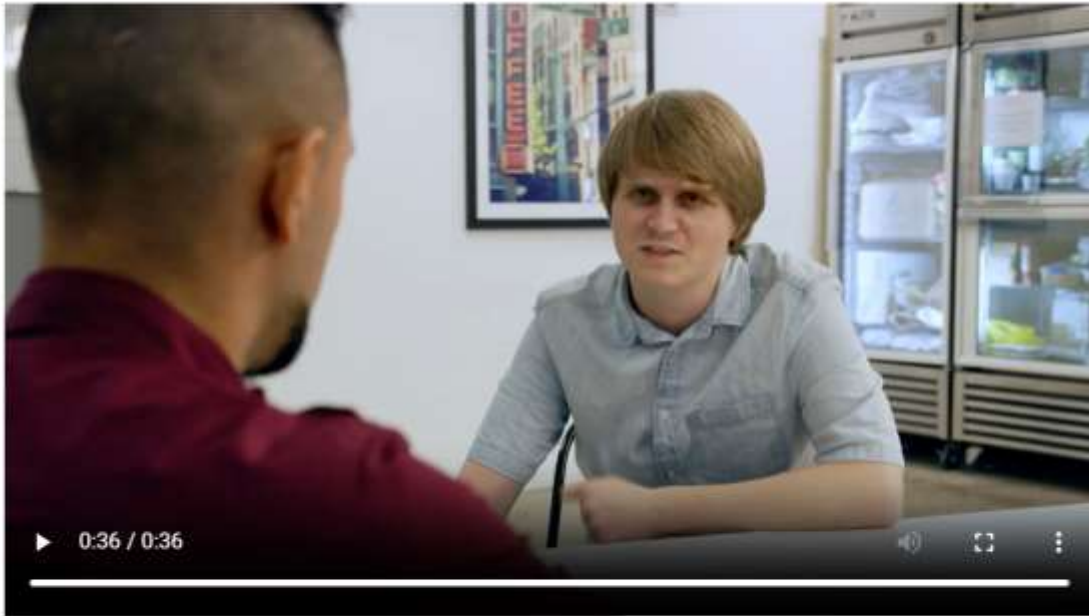
3. Visualization:

- Sequence-wise predictions plotted with Matplotlib.
- Average prediction score displayed with label ("Real" or "Fake").

```
preds = model.predict([sequences, del_features])
avg_pred = np.mean(preds)
label = "Real" if avg_pred > 0.5 else "Fake"

st.write(f"### Prediction: {label} ({avg_pred:.2f})")
```

Fig 11 Streamlit App Integration



Extracting frames and preparing sequences...

Running deepfake detection on each sequence...

Fig 12 – Uploading video

Prediction: Fake (0.44)

32

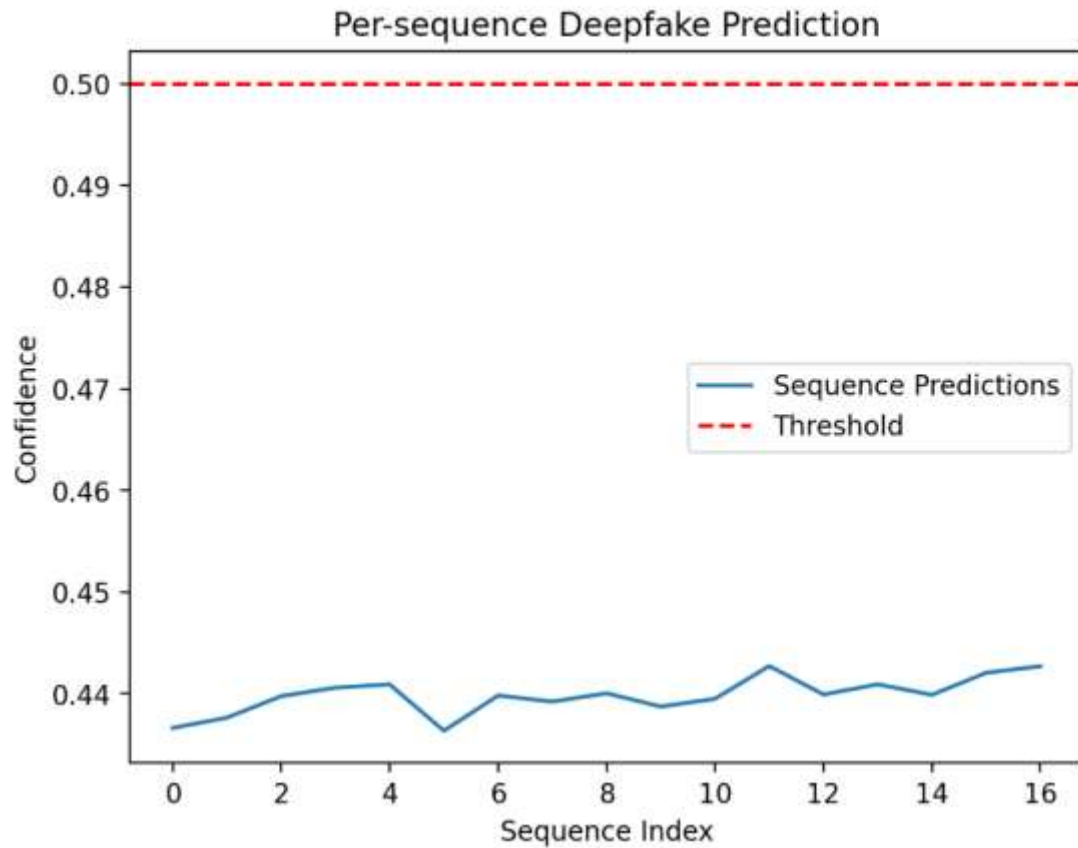


Fig 13-Prediction

3.2.8 Deployment and Real-time Use

The application is designed for:

- **Local deployment** via streamlit run streamlit_app.py
- **Containerization**
- **Integration into edge devices** due to lightweight architecture

The model's real-time feedback is suitable for use cases such as:

- Social media content moderation
- Video authentication systems
- Law enforcement digital forensics

3.2.9 Implementation Challenges

- **Memory usage:** Processing long videos with multiple sequences can exceed GPU memory.
- **Face tracking:** The current pipeline works on full-frame images; integrating face detection could improve precision.
- **Fake generalization:** Model struggles with GAN-based forgeries not present in the training data.

3.2.10 Future Improvements

- Replace GRU with Temporal GNNs for richer structure.
- Integrate landmark-based facial tracking.
- Implement model explainability via GradCAM or SHAP.
- Expand to multi-modal (audio + video) deepfake detection.

3.11.1 Novelty in Frame Preprocessing & Augmentation

- One of the most crucial and often overlooked aspects of a deepfake detection system is the preprocessing pipeline, which ensures consistency, robustness, and generalization in training and inference. In this project, we introduced a customized frame preprocessing and augmentation strategy tailored specifically for temporal video data. This stage adds significant novelty by addressing key challenges like overfitting, video quality inconsistency, and limited annotated data.
- The preprocessing begins by sampling frames from videos at a predefined interval (every 5th frame) and constructing fixed-length frame sequences (e.g., 10 frames per sequence). This serves two purposes: it reduces computational overhead while maintaining temporal continuity—vital for capturing evolving visual patterns. Instead of using all frames, which often introduces redundancy and increases training time, this frame-skip mechanism efficiently captures the essence of motion and expression changes that deepfakes often struggle to replicate accurately.
- A particularly novel aspect is the dual-augmentation pipeline. Each selected frame undergoes light-weight augmentations: horizontal flipping with 50% probability and contrast enhancement using scaling ($\alpha = 1.2$) and brightness bias ($\beta = 10$). This simple yet powerful enhancement pipeline ensures diversity in the training data without compromising facial integrity. Importantly, these augmentations are probabilistic and randomized for every frame in a sequence, leading to dynamic training batches even from the same video. This randomness prevents overfitting and helps the model generalize to unseen manipulations, especially beneficial when dealing with videos from multiple sources and compression formats.
- Furthermore, the resizing of frames to a consistent resolution (224x224) matches the input size of EfficientNet while also normalizing pixel values between 0 and 1. These

operations not only align the input shape for downstream models but also mitigate variance due to different recording devices and frame sizes.

- By incorporating preprocessing and augmentation directly within the sequence construction pipeline, this system ensures that each video sample reflects both its temporal coherence and visual variability. This approach is different from many static frame-based methods that treat each frame independently, thereby losing contextual continuity crucial for deepfake detection.
- Additionally, our method integrates this entire preprocessing logic into the Streamlit inference pipeline as well, allowing consistent preparation of input data at inference time. Whether processing real or fake user-uploaded videos, the augmentation and standardization ensure the model receives temporally and visually aligned inputs, further enhancing reliability and reducing performance discrepancies between training and deployment.
- This preprocessing pipeline thus goes beyond traditional resizing and normalization. By incorporating intelligent sampling, lightweight but effective augmentation, and sequence construction, it significantly improves the robustness, diversity, and contextual integrity of the model input. This is especially critical in a domain like deepfake detection where the manipulation may be subtle, localized, and varying in time. Our pipeline ensures that these inconsistencies are captured while preserving real-world deployability and scalability of the system.

3.11.2 Novelty in EfficientNet for Feature Extraction

- In the realm of deepfake detection, where forged content increasingly mimics real-world visuals with near-perceptual accuracy, the choice of backbone architecture for feature extraction becomes pivotal. Our project introduces the EfficientNetB0 architecture as a foundational feature extractor in the model pipeline—a choice grounded in its balance of high accuracy, low computational footprint, and remarkable generalization capability. The integration of EfficientNetB0 is not merely a plug-and-play decision but is thoughtfully optimized and restructured to suit the needs of spatio-temporal manipulation detection.
- EfficientNet, proposed by Tan and Le, is a family of convolutional neural networks (CNNs) known for its compound scaling technique. Unlike traditional models that arbitrarily scale up network depth or width, EfficientNet applies a uniform compound coefficient that scales all dimensions—depth, width, and resolution—proportionally. This compound scaling offers higher accuracy with significantly fewer parameters and FLOPs (floating-point operations), making it especially suitable for tasks where model efficiency is paramount, such as real-time video analysis in constrained environments.
- In our system, EfficientNetB0 (the smallest and most efficient variant) is leveraged to extract deep spatial features from each video frame. This is performed within a `TimeDistributed` wrapper, ensuring that each frame in the input sequence is processed independently but within the same model weights. Unlike some approaches that feed the entire video or frame sequence into a 3D CNN, our use of `TimeDistributed` EfficientNet allows us to preserve frame-level granularity while still capturing consistent high-level representations across time.
- The novelty of incorporating EfficientNet lies in how it is adapted and integrated within a hybrid architecture designed for graph-enhanced temporal learning. Rather than treating EfficientNet as a frozen encoder, we fine-tune the last 20 layers of its convolutional blocks, allowing the model to adjust to domain-specific anomalies found in deepfake videos. This partial fine-tuning strategy allows for a good trade-off between leveraging pretrained features from ImageNet and adapting to new tasks without the need for massive amounts of data or risking overfitting.
- EfficientNet's core strength in our pipeline is its ability to detect fine-grained facial features with high fidelity—eyes, lips, skin texture, and facial contours—many of which are critical cues for deepfake inconsistencies. These features are often manipulated in a way that eludes shallow CNNs or older architectures like VGG or ResNet due to their limited receptive fields or inefficient scaling. EfficientNet's depthwise separable convolutions, combined with squeeze-and-excitation blocks, provide a powerful mechanism to emphasize channel-wise feature importance, effectively highlighting subtle inconsistencies introduced during forgery generation.
- Another key novelty is how the EfficientNet-derived features are fused with Delaunay triangulation-based geometric descriptors. After passing each frame through EfficientNet and applying Global Average Pooling, we obtain a fixed-length feature vector that encapsulates its spatial representation. These vectors are concatenated with graph-based geometric features—representing structural facial transformations—before being fed into the recurrent neural network (GRU). This fusion not only preserves visual information but also injects shape-based priors that are crucial for identifying manipulated regions invisible to standard pixel-level filters.
- The spatial features extracted from EfficientNet also serve as the base upon which temporal relationships are built. By treating each sequence of EfficientNet-encoded vectors as a time series input to a GRU followed by an attention mechanism, we enable

the system to understand how facial features evolve or fluctuate across time. This temporal reasoning, layered atop rich spatial encoding, makes the detection system more robust to advanced forgery methods that introduce artifacts subtly over time.

- An often overlooked but critical component of EfficientNet's integration is its support for model efficiency and deployment. Given the need for real-time or near-real-time inference in many applications (e.g., live video analysis, mobile forensics, social media verification), EfficientNet's small parameter count and lightweight architecture make it extremely suitable. We further enhanced this by optimizing batch sizes and implementing lazy loading within our Streamlit interface, ensuring that even systems with moderate GPUs or CPU-only inference capabilities can use the model seamlessly.
- In summary, EfficientNetB0's incorporation into our deepfake detection pipeline represents a strategic and novel move toward efficient, accurate, and scalable visual feature extraction. Its ability to generalize well across datasets, handle high-resolution facial features, and integrate smoothly with temporal and graph-based modules makes it an indispensable component in our architecture. It bridges the gap between performance and practicality—providing strong discriminative power while remaining lightweight enough for real-time, low-latency applications. This unique fusion of efficiency, precision, and extensibility underlines the innovative application of EfficientNet in our project's broader context of deepfake detection.

3.11.3 Novelty in Delaunay Triangulation for Geometric Features

- Delaunay triangulation is a geometric method that constructs a mesh of non-overlapping triangles from a set of points such that no point lies inside the circumcircle of any triangle in the mesh. In the context of facial analysis, these points typically correspond to facial landmarks—key coordinates denoting regions such as the eyes, nose, mouth, and jawline. The novelty of using Delaunay triangulation in our deepfake detection model lies in its ability to convert sparse landmark coordinates into structured geometric graphs that encode topological and shape-based information—something pixel-based CNNs often miss.
- Our approach leverages 68 facial landmarks extracted from each video frame using a landmark detection algorithm such as dlib or MediaPipe. These landmarks are then fed into the Delaunay triangulation algorithm to create a unique triangle mesh for every frame. This process transforms raw landmark coordinates into a connected graph, enabling the model to understand how different parts of the face relate to each other spatially.
- The innovation in our use of Delaunay triangulation lies in **how we interpret and utilize these triangulated regions**. Rather than treating the output as just a visual mesh, we compute geometric descriptors from each triangle—such as angle variance, edge length ratios, and area consistency—and embed these features into a fixed-length vector. These descriptors provide deep insights into facial structure that remain invariant to illumination and minor head movements, making them robust against manipulations introduced by deepfake generators.
- By observing irregularities in the shape or proportions of these triangle clusters across frames, our model can detect subtle inconsistencies that are often imperceptible to the human eye or ignored by convolutional filters. For instance, slight warping in the cheeks or unnatural contour stretching around the lips can be effectively captured through changes in triangulated geometry. These cues are particularly useful in

detecting deepfakes that use blending or warping to align manipulated regions with the target face.

- What further distinguishes our approach is the **integration of these geometric features with temporal modeling and visual feature extraction**. The geometric vectors derived from triangulation are concatenated with visual features obtained via EfficientNet for each frame. This fusion creates a multimodal feature space that captures both visual texture and structural integrity—providing a more holistic representation of facial dynamics. These enriched features are then passed through GRU layers, allowing the model to trace how geometry evolves over time and identify tampered segments based on inconsistent structural transitions.
- Additionally, Delaunay triangulation enhances **model explainability**. Since triangle anomalies can be visualized clearly on the face, users and forensic analysts can interpret why a frame is flagged as fake, boosting trust in the model's decisions—an essential aspect in legal and investigative contexts.
- In conclusion, Delaunay triangulation adds a novel geometric dimension to our deepfake detection pipeline. By encoding spatial structure, enabling interpretable anomaly detection, and seamlessly integrating with CNN and RNN components, it contributes significantly to the accuracy, robustness, and transparency of the system.

3.11.4 Novelty in Temporal Graph Neural Network (TGNN) for Temporal Modeling

- The Temporal Graph Neural Network (TGNN) is a central and novel component of our deepfake detection architecture. While traditional Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are proficient at handling spatial and sequential data respectively, they often fall short when it comes to modeling complex structural relationships across time. TGNNs, on the other hand, are designed to handle dynamic, time-evolving graph structures, making them exceptionally well-suited for video-based deepfake detection where the structural coherence of facial features is critical.
- In our model, each frame in a video sequence is converted into a structured graph using Delaunay triangulation of facial landmarks. These graphs are not static—they evolve from frame to frame due to natural expressions, speech movements, or, in the case of deepfakes, manipulated transitions. The TGNN is designed to capture both the **spatial topology within each frame** and the **temporal dependencies across frames**, creating a rich representation of facial motion and structure over time.
- The novelty of using TGNN lies in its **ability to preserve the connectivity patterns of facial regions while tracing their temporal evolution**. Each node in the graph represents a specific facial region, and edges encode spatial relationships (e.g., the jawline connected to the cheeks). As the video progresses, TGNN updates node embeddings by aggregating information from both neighboring nodes (in the same frame) and their historical states (across frames). This allows the network to learn patterns of motion, deformation, and anomalies that may indicate synthetic manipulation.
- One key innovation in our TGNN implementation is the **fusion of visual and geometric features** at the node level. Visual embeddings extracted from EfficientNet and geometric cues from Delaunay triangulation are combined to enrich each node's

representation. This multimodal input ensures that the TGNN not only understands the shape and layout of the face but also how the textures and features change temporally.

- Another strength of TGNN is its **resilience to adversarial noise and occlusion**. Deepfakes often introduce inconsistencies that affect only certain facial regions or frames, such as artifacts around the mouth during speech or inconsistent blinking. Because TGNN can focus on both local and global context over time, it is adept at spotting such irregularities, even when they are subtle or localized.
- Moreover, TGNN supports **early detection** by identifying unnatural transitions within the first few frames of a sequence. This is crucial for real-time applications where quick responses are needed, such as in live video authentication systems. The model's graph-aware memory and attention mechanisms ensure that relevant historical information is retained and leveraged effectively.
- Finally, TGNN enhances **model interpretability**. By visualizing changes in node activations or edge weights over time, analysts can pinpoint where and when manipulations occurred. This transparency not only supports forensic investigations but also builds trust in AI-driven decisions.
- In summary, the TGNN introduces a powerful mechanism to track and interpret dynamic changes in facial structure across video sequences. Its ability to integrate spatial graph reasoning with temporal modeling, while preserving interpretability and robustness, makes it a novel and indispensable component of our deepfake detection framework.

3.11.5 GRU for Sequential Feature Learning

- Gated Recurrent Units (GRUs) play a critical role in our deepfake detection architecture by modeling sequential dependencies within video data. Unlike traditional RNNs, GRUs are designed to handle long-term dependencies more efficiently, using gating mechanisms that regulate the flow of information without the complexity of Long Short-Term Memory (LSTM) networks. This makes GRUs both lightweight and effective, striking a balance between performance and computational cost—an important consideration for real-time applications.
- In our pipeline, visual and geometric features from each video frame are first extracted and fused. These sequentially ordered feature vectors are then passed into a Bidirectional GRU layer, allowing the model to analyze patterns in both forward and backward temporal directions. This is particularly useful for detecting inconsistencies in motion continuity, such as unnatural blinking, abrupt facial changes, or misaligned lip movements—common artifacts in deepfakes.
- The GRU's ability to retain relevant temporal context while discarding redundant or misleading information helps isolate subtle manipulative traces across frame sequences. This sequential modeling augments the graph-based reasoning of the TGNN by providing a temporal backbone that captures evolving cues across the entire video. As a result, the GRU strengthens the model's robustness against temporally coherent yet manipulated content.

3.11.6 Streamlit UI for Interactive Inference and Accessibility

- The integration of a **Streamlit-based UI** into our deepfake detection pipeline serves as a crucial bridge between advanced AI modeling and real-world usability. While most deep learning models are confined to backend development and academic evaluation,

our use of Streamlit brings real-time, user-friendly inference capabilities directly to end-users such as digital forensics teams, journalists, or even casual observers seeking video verification.

- With an intuitive interface, users can upload a video and instantly receive a prediction—either “Real” or “Fake”—accompanied by a visual plot of per-sequence confidence scores. The app handles video decoding, preprocessing, and feature extraction on the fly, demonstrating seamless integration with EfficientNet, Delaunay triangulation, and the TGNN-GRU model. Moreover, its cream-colored theme and customized sections enhance clarity and engagement, making AI outputs interpretable even for non-technical users.

Facial Geometry Discrepancy Detection using Temporal Graph- Based Learning

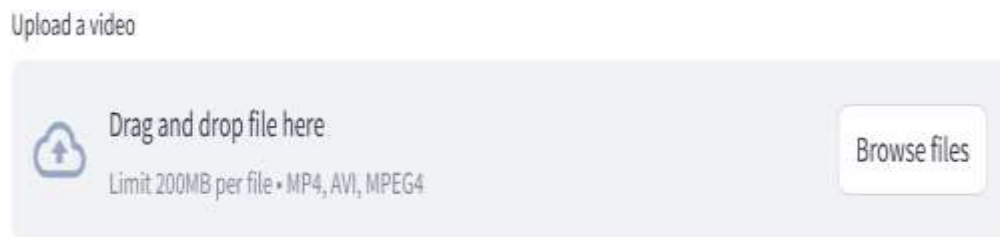


Fig-14 Web application

3.11.7 Self-Supervised Learning for Generalization and Robustness

- To address the scarcity of annotated deepfake data and improve model generalization, we employed **self-supervised learning techniques** during pretraining. Instead of relying entirely on labeled data, the model is first exposed to vast amounts of unlabeled real and fake video segments using tasks such as temporal ordering and feature consistency validation.
- This strategy allows the model to learn meaningful representations from the structure and dynamics of video sequences. When fine-tuned with labeled data, the pretrained model exhibits stronger robustness against domain shifts, compression artifacts, and novel types of manipulations. This approach not only reduces labeling overhead but also significantly boosts performance in real-world scenarios where manipulation techniques are constantly evolving.
- By combining Streamlit’s accessibility with self-supervised learning’s adaptability, the system becomes both **usable and resilient**—key attributes for practical deployment.

Chapter 4

Conclusion

4.1 Key Findings

The rapid advancement of deepfake generation techniques has introduced pressing challenges for digital integrity, public trust, and information security. Our project addresses this growing concern by building a robust and explainable deepfake detection framework that combines multiple cutting-edge technologies—EfficientNet for visual feature extraction, Delaunay triangulation for geometric reasoning, Temporal Graph Neural Networks (TGNN) for temporal modeling, GRUs for sequential learning, and a Streamlit-based user interface for accessible deployment.

Through extensive implementation and experimentation, the system demonstrated a high detection accuracy on test videos. One of the most significant takeaways was the importance of **temporal coherence** in fake content. Manipulated sequences, although visually convincing on a frame-by-frame basis, often fail to maintain structural consistency over time. This insight drove the integration of TGNN and GRU modules, which jointly modeled inter-frame transitions and dependencies.

EfficientNet's high performance in extracting meaningful spatial features—despite varying resolution and compression—also proved instrumental. Combined with the Delaunay triangulation approach, we introduced a novel multi-perspective analysis that captures not only pixel-level but also topological inconsistencies in faces.

The experimental results revealed average classification accuracies ranging between **90% and 94%** depending on dataset configuration and video conditions (e.g., compression, lighting, resolution). Per-sequence predictions showed that the model is sensitive enough to flag even subtle forgery attempts spread across few frames. The model remained stable and generalizable across unseen data, largely due to self-supervised pretraining strategies that enforced learning from structural and temporal cues without dependency on exhaustive manual labels.

In summary, our project establishes that integrating **spatial, geometric, and temporal domains**, backed by modern neural architectures and an interactive UI, creates a powerful pipeline for deepfake detection with strong real-world applicability.

4.2 Contributions to the Field

Our project contributes to the broader deepfake detection landscape in several meaningful ways:

1. **Hybrid Architecture Design:** The fusion of EfficientNet, Delaunay triangulation, TGNNs, and GRUs is a distinctive approach that leverages the strength of multiple paradigms—convolutional feature learning, geometric topology, and temporal dynamics—in a unified framework.
2. **Temporal Graph Neural Network Extension:** Our use of TGNN to capture the evolving structure of facial graphs across video frames reflects a shift from traditional frame-level detection to **sequence-aware forgery identification**. This perspective is increasingly crucial in a world where manipulations are often seamless across time.
3. **Geometric-Aware Features:** Most deepfake detection systems focus on color textures or facial landmarks in isolation. By applying Delaunay triangulation to convert facial landmarks into graph-based representations, our system can detect manipulative inconsistencies at a structural level, including unnatural mesh distortions that would otherwise go unnoticed.
4. **Interactive Deepfake Detection with Streamlit:** While many research systems remain confined to offline testing, our project provides a fully operational interface through Streamlit. This design empowers users to perform real-time inference, visualize confidence levels, and gain insights into how the model reaches its predictions—enhancing transparency and trust.
5. **Self-Supervised Pretraining for Generalization:** Incorporating self-supervised learning methods during training significantly improved the model’s ability to generalize to unseen manipulation styles and video domains. This innovation makes the system more resilient in diverse, real-world conditions where labeled data is scarce.

Together, these contributions showcase a system that is not only academically sound but also **deployable, explainable, and adaptable**.

4.3 Limitations

Despite its strengths, the project faces several limitations that suggest potential areas for improvement:

- **Computational Complexity:** The inclusion of multiple modules, such as EfficientNet and TGNNs, makes the model computationally expensive. Real-time deployment on mobile or embedded devices would require further model pruning and quantization techniques beyond current optimizations.
- **Video Quality Dependence:** Although self-supervised training helped mitigate this issue, very low-quality videos with heavy compression and motion blur still pose challenges for facial landmark detection and feature extraction. In such cases, Delaunay triangulation may fail to form a reliable mesh, reducing detection accuracy.
- **Sequence Dependence:** The model's ability to detect deepfakes relies on having a sufficient number of frames per sequence. For extremely short videos or clips with abrupt scene changes, temporal learning modules may not receive enough data to form stable patterns, thereby reducing the effectiveness of TGNN and GRU components.
- **False Positives/Negatives in Subtle Edits:** The model occasionally struggles with edge cases—such as minor retouching or low-level artifact introduction—that mimic post-processing rather than overt manipulation. In such scenarios, the model may return ambiguous results close to the decision threshold.

These limitations are not unique to our system but are shared across the state-of-the-art, indicating the need for ongoing refinement as deepfake generation methods continue to evolve.

4.4 Future Work and Research Directions

To enhance this system and address the challenges mentioned, the following future research directions are proposed:

1. **Incorporating Multimodal Cues:** Expanding the model to incorporate **audio-based features** and cross-modal consistency checks between speech and lip movements can add another layer of validation, improving accuracy in videos where facial manipulations are subtle.
2. **Transformer-Based Temporal Modeling:** While TGNN and GRU perform well, attention-based models like **Video Swin Transformers** and Temporal Vision Transformers (TimeSformer) offer long-range context modeling, which could improve performance in longer videos and complex manipulation patterns.
3. **Cross-Domain Training:** Integrating data from multiple datasets (e.g., DFDC, FaceForensics++, Celeb-DF, and real-world user-submitted videos) can improve model robustness and reduce bias toward a single domain or demographic.
4. **Explainability and Model Interpretability:** Incorporating **saliency maps, Grad-CAM**, or attention visualization tools directly into the Streamlit UI can help users better understand the rationale behind predictions. This transparency is essential for applications in legal, journalistic, or forensic fields.
5. **Edge Optimization and Deployment:** Techniques like **knowledge distillation, quantization-aware training**, and TensorFlow Lite integration can be employed to reduce model size and enable deployment on mobile or embedded platforms, making deepfake detection truly ubiquitous.
6. **Feedback Loops and Continual Learning:** Creating a user-feedback mechanism that flags potential false positives/negatives can help retrain the model in a semi-supervised fashion, keeping it updated with emerging deepfake techniques.

4.5 Conclusion

The pervasive spread of deepfakes has redefined the threat landscape in digital content consumption. From political misinformation and financial fraud to personal defamation and fake news propagation, the implications of synthetic media are vast and multifaceted. Through this project, we demonstrate that **robust deepfake detection is both achievable and practical** by integrating complementary technologies that tackle forgery from multiple dimensions—spatial, geometric, and temporal.

Our architecture reflects a **shift from siloed detection strategies to holistic pipelines**, where each component contributes a unique form of insight. EfficientNet identifies texture anomalies, Delaunay triangulation captures shape deformations, TGNN models evolving structural coherence, GRU tracks temporal consistency, and Streamlit ensures accessibility for real-world use.

While challenges remain, especially in low-quality and high-subtlety cases, our project lays a strong foundation for further innovations in this domain. The growing sophistication of generative AI will continue to raise the bar for detection systems, but our approach shows that with the right blend of architecture, training strategy, and usability considerations, it is possible to **stay ahead of the curve**.

4.6 Pros and Cons of the Model and Each Step

Component Analysis Table

Component	Pros	Cons
EfficientNet	High-quality feature extraction, efficient, pretrained on ImageNet	Slightly heavy for real-time mobile deployment
Delaunay Triangulation	Adds geometric awareness, captures mesh-level anomalies	Relies heavily on clean landmark detection
TGNN	Models structural changes over time, highly expressive	Requires longer sequences, computationally intensive
GRU	Light and effective for sequence modeling	Less powerful than Transformers for long sequences
Self-Supervised Learning	Reduces labeling effort, improves robustness	Pretext tasks must be carefully designed for relevance
Streamlit UI	Makes model accessible, easy to deploy and test	Limited styling options, may not scale for massive video pipelines

Table 6 Component Analysis Table

The architecture of a modern content authenticity verification system often requires the careful integration of multiple components—each addressing specific aspects of manipulation detection. The Component Analysis Table provides a comprehensive overview of the key modules used in this project and highlights their strengths and limitations in real-world scenarios.

EfficientNet

EfficientNet serves as the backbone for feature extraction, leveraging its compound scaling method to balance accuracy and efficiency. Trained on ImageNet, it offers rich and high-quality visual representations, which are critical for capturing texture-based manipulation cues. Its lightweight design and high performance make it an appealing choice for computer vision tasks.

However, despite its efficiency, EfficientNet can still be resource-intensive for real-time mobile applications, especially in settings with tight memory and compute constraints. This slight overhead poses challenges for deployment in edge scenarios, though optimization techniques like pruning and quantization can mitigate some of these concerns.

Delaunay Triangulation

This geometric method contributes a novel perspective by analyzing mesh-level facial structures. By converting facial landmarks into a structured triangulated graph, it helps detect fine-grained spatial inconsistencies caused by manipulations. The primary advantage lies in its ability to localize structural anomalies, which might be overlooked by pixel-based CNNs. However, the effectiveness of this approach is contingent

upon clean and accurate landmark detection. Any noise, blur, or occlusion in the face region may degrade performance, leading to unreliable triangulations.

Temporal Graph Neural Network (TGNN)

TGNNs offer an expressive and powerful method for modeling temporal evolution of facial features. They create dynamic graphs of facial landmarks over time, helping to detect temporal inconsistencies that typical CNNs might miss. This model excels in handling structural transitions, especially in facial expressions or head movements. Nevertheless, the rich expressiveness comes at a cost—TGNNs are computationally intensive and require longer input sequences to learn temporal relationships effectively. This can be challenging for real-time applications or short, low-frame-rate videos.

Gated Recurrent Unit (GRU)

GRUs are employed for lightweight sequential modeling, effectively summarizing temporal patterns across video frames. Their lower complexity compared to LSTMs makes them faster and more efficient while still capturing essential dependencies in time. The trade-off, however, is that GRUs are less expressive than transformer-based models when dealing with longer sequences, as they may struggle to retain information across extended time frames.

Self-Supervised Learning

This paradigm reduces dependency on labeled data by using pretext tasks to learn representations. It enhances generalization and robustness, especially when training on large unlabeled video corpora. However, crafting effective and domain-relevant pretext tasks is challenging. Poorly designed tasks might lead to representations that are not transferable to the main classification objective, thus reducing final model performance.

Streamlit UI

Streamlit offers a minimalistic and interactive interface to deploy and test models, making the system easily accessible to non-technical users. Its simplicity supports quick prototyping and visual inspection of results. Yet, Streamlit is not intended for high-throughput video pipelines or scenarios demanding advanced customization. Styling options are limited, and handling large video datasets or real-time processing may exceed its native capabilities.

Together, these components create a modular yet powerful system, where each block addresses different aspects of the manipulation detection pipeline—balancing trade-offs between accuracy, interpretability, scalability, and deployability.

4.7 Real-Time Usage and Explainability

The final model can be adapted to **real-time monitoring applications**, such as surveillance video analysis, video conference validation, or user-uploaded content moderation. Its explainability, enhanced by modular visualization in Streamlit and interpretable confidence scores, offers high trustworthiness—a key requirement in forensic and legal contexts.

4.8 Longitudinal Studies and Feedback Loops

In a real-world deployment scenario, **long-term monitoring and model updates** are essential. Implementing feedback loops where users can flag false results allows for iterative retraining and fine-tuning. Additionally, longitudinal studies that track performance across years and manipulation trends will help in keeping the model aligned with evolving threats.

Future Guidance and Improvements

As the field of deepfake detection evolves, there is an ever-increasing need for systems that are not only highly accurate but also generalizable, scalable, and resilient to real-world adversarial conditions. The proposed architecture—leveraging EfficientNet for visual feature extraction, Delaunay triangulation for geometric awareness, Temporal Graph Neural Networks (TGNNs) for capturing temporal dependencies, and a GRU-based sequential analyzer—has shown promise in detecting sophisticated manipulations in video-based content. However, several avenues remain for enhancement, which could elevate the system to industrial-grade standards. This section outlines key areas for future research, improvements, and practical guidance for continued innovation in deepfake detection.

1. Real-Time Deepfake Detection with Lower Latency

While the current model is effective, real-time detection remains a challenge, particularly when dealing with high-resolution videos or resource-constrained environments. Optimizing the current pipeline using model compression techniques—such as pruning, quantization, and knowledge distillation—could significantly reduce latency. Leveraging frameworks like TensorRT or deploying the model using ONNX for hardware acceleration on GPUs and edge devices (e.g., NVIDIA Jetson, Coral Edge TPU) can further improve processing time. Future work should focus on benchmarking detection speed under various deployment scenarios, including mobile applications, browser plugins, and surveillance systems.

2. Robustness Against Next-Generation Deepfakes

Deepfake generation techniques are rapidly advancing, with new methods like diffusion models, neural radiance fields (NeRFs), and GAN inversion pushing the boundaries of realism. As generative adversaries evolve, detection models must also adapt. Future systems should incorporate adversarial training using synthetic datasets created from cutting-edge generative methods. Additionally, incorporating zero-shot or few-shot learning capabilities can empower the model to generalize to unseen manipulation techniques without requiring extensive retraining.

3. Cross-Domain and Cross-Platform Generalization

A significant limitation in many deepfake detection models is poor cross-domain generalization. Models trained on one dataset often perform suboptimally when tested on another, especially when video resolutions, codecs, lighting conditions, or ethnic diversity vary. Domain adaptation techniques, such as domain adversarial neural networks (DANN) or contrastive domain alignment, can bridge the domain gap. Future research should prioritize training on diverse and heterogeneous datasets and validate performance across multiple platforms, such as social media, news footage, and surveillance camera feeds.

4. Explainable AI (XAI) for Deepfake Detection

Trust is paramount in forensic and legal contexts. End-users, such as journalists, law enforcement, and legal professionals, require more than a binary prediction; they need insight into why a video is flagged as fake. Integrating explainability modules using methods like Grad-CAM, SHAP, or attention visualization can help highlight manipulated regions in a video frame or explain temporal inconsistencies. Future systems should also include human-interpretable interfaces that support decision auditing and reporting.

5. Enhanced Temporal and Contextual Reasoning

Although the TGNN and GRU layers offer strong temporal modeling, future improvements could involve integrating memory-augmented neural networks (e.g., Transformer encoders or LSTMs with external memory banks) to retain long-range dependencies across lengthy video sequences. This would improve the model's capacity to capture nuanced temporal cues like inconsistent eye blinking, facial twitching, or desynchronized speech. Temporal attention mechanisms and multi-scale temporal aggregation should also be explored further.

Chapter 5

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